

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-a39117b9cab043c1c.jsonl`
- SHA-256: `77a6f922bc04a330967341aa0f6f71aad0958ce2917d242f4f8dc72ea1b85ae5`
- Source modified: `2026-06-22T01:58:18+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf_6abc9156-811`
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T01:55:49.367Z)

You are scanning planning docs to surface ACCURATE, prioritized next steps for a co-founder.

REPO RULES:

```
- Engine repo at /c/Users/avidu/Projects/badminton-highlight-indexer: its working tree is on a
STALE serving branch. Read DOCS + verify SHIPPED state from ENGINE origin/master ONLY:
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer show origin/master:docs/FILE.md
(read a doc on master; if it errors, the doc may be branch-local ? fall back to the working-tree
file and SAY SO)
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60
(what actually shipped ? use this to catch stale "TODO/next" claims)
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "PATTERN" origin/master
-- 'PATHGLOB'
- khelsutra repo at /c/Users/avidu/Projects/khelsutra: on clean master ? use Read/Grep/Glob
directly + git -C /c/Users/avidu/Projects/khelsutra log --oneline -40.
```

ANTI-SPIRAL (critical): before listing any "[gap]/net-new/TODO/next" item, VERIFY it hasn't already shipped ? cross-check the doc's claim against git log origin/master. A doc saying "X is next" is a HYPOTHESIS until you confirm X isn't already merged. Flag any stale doc claims you correct.

Context (already TRUE as of this session, do not re-list as "next"): the khelsutra ALPHA is DEPLOYED + live on a CPU droplet (engine + cloudflared tunnel + Cloudflare Access + SPA on Pages, real Gemini key working, full chain verified). DEPLOY_ALPHA fixes + the droplet kit are merged (khelsutra #78/#79/#81/#85). T2 (alpha deploy) is effectively DONE.

Return prioritized next items with honest status + git grounding.

```
TRACK: **khelsutra UX / localization / runnable-tool**. Read
/c/Users/avidu/Projects/khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md ($7 tracker + $9 DoD),
/c/Users/avidu/Projects/khelsutra/docs/MULTILINGUAL_EXPANSION.md ($8 owner Qs),
/c/Users/avidu/Projects/khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md ($9 T-series). Cross-
check A-items + T-items against `git -C /c/Users/avidu/Projects/khelsutra log --oneline -40`.
ANSWER: current status (A-items, the 6-lang SPA, T2 deploy now done); the genuinely-open next
items ? the **native-speaker review** (which catalogs remain: es/da/id + watermark? kn/hi
console did #77), **MULTILINGUAL Phase B (Thai)/C (CJK)** (owner-gated $8 Q2-Q5), any remaining
T/A items. Prioritize.
```

2. assistant (2026-06-22T01:55:52.391Z)

I'll start by pulling the latest and reading the three target docs, then cross-check against git log.

3. user (2026-06-22T01:55:57.101Z)

Already up to date.

---LOG---

17242a9 feat(site): localize own-model body content to kn/hi (#86)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix) (#85)
5e5a2ac docs(friendly-multilingual): close out the chapter (\$9 DoD + UI/UX + B7 deferred) (#84)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero (#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit (#80)
8328721 docs(tracker): refresh the \$7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook (RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)
f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru (#71)
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling tracked (#301) (#69)
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.* tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin /api) (#64)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*) (#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
ccdcfcd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)
a0475bc docs: hygiene refresh ? sync trackers to shipped reality (#59)
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD \$6 box 3 ?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download, multilingual) (#53)
1b69e1a feat(site): show a localized UI screenshot on each academy flyer (#52)
3a42847 feat(web): localized sample-query prompt (en/kn/hi) + i18n-test rule for query parsing (#51)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)
631c3aa docs(friendly-multilingual): SPA fully trilingual ? mark B1/B2/B4/B5/B6 done + screencast milestone (#49)
ad0d7ed feat(web): ship the Hindi (hi) catalog ? tracker B6 (the extensibility proof) (#48)
4449981 feat(web): ship the Kannada (kn) catalog ? tracker B5 (#47)
a423b82 feat(web): self-host OFL Noto Kannada + Devanagari fonts ? tracker B4 (#46)

4. user (2026-06-22T01:55:58.928Z)

fetchcd

5. assistant (2026-06-22T01:56:02.770Z)

Good, the khelsutra repo is current. Now let me read the three target docs.

6. user (2026-06-22T01:56:03.680Z)

```
1      # Multilingual Expansion ? 7 new locales (th · id · da · es · zh-Hans · ja · ko)
2
3      > **Status:** `IN PROGRESS` ? **Phase A (es/da/id) shipped
([#67](https://github.com/avidullu/khelsutra/pull/67)); Phases B (Thai) / C (CJK) owner-gated**
· **Owner:** Avi Dullu · **Created:** 2026-06-21 · **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; index in
`docs/README.md`; pointer in memory `current-state`; thread in `NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** **PEER-OF / extends**
[FRIENDLY_AND_MULTILINGUAL.md](FRIENDLY_AND_MULTILINGUAL.md) (the `en`/`kn`/`hi` foundation)
and the engine design `I18N_PLAN.md`. **Reuses that foundation verbatim** (react-i18next +
i18next-icu + browser detector + key-parity/no-bare-strings CI + self-hosted OFL Noto via
`@fontsource` + Gemini-draft?native-review). This is a **parallel** line: it adds locales, it
does **not** re-architect i18n.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs the working tree this
session), `[researched]` (sourced, needs sign-off), or `[design]` (proposed). Not legal advice ?
the OFL-font carve-out is the owner's standing call (D7 in FRIENDLY).
8
9      ---
10
11     ## 0. TL;DR
12     Add **Thai, Indonesian, Danish, Spanish, Chinese (Simplified), Japanese, Korean** to the
Studio SPA with the **same rigor as en/kn/hi**. The good news `[verified]`: the i18n foundation
already scales ? **zero changes to the negotiation contract or the catalog format**; adding a
locale is "drop a reviewed `common.json` + (if non-Latin) bundle one OFL Noto font + keep CI
key-parity green." The work splits cleanly into **Phase A** (es/da/id ? *catalog-only, no
fonts*), **Phase B** (Thai ? *one small OFL font*), and **Phase C** (zh-Hans/ja/ko ? *large OFL
CJK fonts + lazy-load + an offline-bundle subsetting decision + the zh script-variant call*).
**The single most important caveats:** (1) **native-speaker review per language** is the real
cost and the public-launch gate (Gemini drafts are *inference-only*, never the source of truth ?
the standing guardrail); (2) **CJK fonts add tens of MB** to the otherwise-tiny air-gapped
bundle; (3) **Spanish now has a 3-form plural** (`one, many, other`) ? the one authoring
wrinkle.
13
14     ## 1. Problem & goal
15     **Why now.** en/kn/hi shipped and proved the foundation is locale-pluggable (a Kannada-
default browser auto-renders Kannada; the switcher walks kn?hi?en). The owner wants to broaden
reach beyond the India beachhead ? **SE Asia (th, id), Europe (da, es), and East Asia (zh, ja,
ko)**.
16     **Concrete outcome ("good"): a user whose browser is in any of the 7 languages lands
on the Studio (and, once `site/` is localized via FRIENDLY B3, the marketing pages too), and ?
without help ? understands the product, runs the tool, and gets a reel, **entirely in their
language**, with the fonts rendering crisply offline.
17     **Read to get current:** this doc §3?§4;
[FRIENDLY_AND_MULTILINGUAL.md](FRIENDLY_AND_MULTILINGUAL.md) (the foundation this rides on,
esp. §4 design + §6.5 negotiation tests); `web/src/i18n/index.ts` (the detector/ICU/fallback
config) + `web/src/main.tsx` (the `@fontsource` font-load pattern) +
`web/src/i18n/hygiene.test.ts` (key-parity + no-bare-strings + offline guards).
18
19     ## 2. Decisions locked (inherited from FRIENDLY unless noted)
20     | # | Decision | Source / date | Implication |
21     |---|-----|-----|-----|
22     | ED1 | **Same framework, no re-architecture** ? react-i18next + i18next-icu + `i18next-
browser-languagedetector`; one `en` source catalog + per-locale siblings; **always fall back to
`en`** | FRIENDLY D-* / `i18n/index.ts` `[verified]` | Each new locale = add to
`SUPPORTED_LOCALES` + a `common.json`; negotiation/format unchanged |
23     | ED2 | **Self-host OFL Noto via `@fontsource`** (the D7 OFL-for-media carve-out) ?
**extend it to the 4 new Noto fonts** (Thai, SC, JP, KR) | FRIENDLY D7 (owner, 2026-06-21) | No
```

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new licensing decision; same `main.tsx` import + `unicode-range`-scoped pattern as kn/hi |
24 | ED3 | **Gemini-draft ? native-speaker review** (inference only, never training; every
value flagged for review) | `CLAUDE.md` guardrail | Translation *quality* is human-gated +
owner-paced; CI proves parity, not quality |
25 | ED4 | **CI key-parity + no-bare-strings stay merge gates** for every locale |
`hygiene.test.ts` `[verified]` | A new catalog must match the `en` key set exactly or CI fails |
26 | ED5 | **All 7 are LTR** ? no RTL/bidi layout work in this batch | `[researched]`
CLDR/W3C | No `dir="rtl"`, no logical-property audit; RTL (ar/he/ur) would be a separate, larger
track |
27 | ED6 *(proposed)* | **Chinese = `zh-Hans` (Simplified) for v1**; `zh-Hant`
(Traditional) is a *separate* catalog + font, added later only on demand | §8 Q2 `[design]` |
Key the catalog `zh-Hans` (not bare `zh`) so `zh-Hant` slots in cleanly |
28
29 ## 3. Foundation ? research / prior art `[researched]`
30 Per-language diligence (Unicode CLDR plural-rules chart; SIL OFL; `@fontsource`; W3C
BCP-47 ? sources in §10):
31
32 | Lang | BCP-47 | CLDR **cardinal** plurals | Dir | Dedicated font (OFL) |
`@fontsource(-variable)` pkg | Font size / notes |
33 |---|---|---|---|---|---|---|
34 | Thai | `th` | **`other`** (1 form) | LTR | **Noto Sans Thai** | `noto-sans-thai` |
**small** (single script block) · no-space word-wrap gotcha |
35 | Indonesian | `id` | **`other`** (1 form) | LTR | ? (Latin) | ? | pure catalog add |
36 | Danish | `da` | **`one, other`** (2) | LTR | ? (Latin) | ? | same shape as English (`æ
ø å` in Latin font) |
37 | Spanish | `es` | **`one, many, other`** (3) ? | LTR | ? (Latin) | ? | `many` =
millions/compact only; ~15?30% string expansion |
38 | Chinese (Simpl.) | **`zh-Hans`** | **`other`** (1 form) | LTR | **Noto Sans SC** |
`noto-sans-sc` | **very large** (~7?16 MB/weight unsubset) · script-variant decision |
39 | Japanese | `ja` | **`other`** (1 form) | LTR | **Noto Sans JP** | `noto-sans-jp` |
**very large** (CJK) |
40 | Korean | `ko` | **`other`** (1 form) | LTR | **Noto Sans KR** | `noto-sans-kr` |
**very large** (Hangul + Han) |
41
42 **Three findings that drive the design:**
43 1. **Spanish's 3-form plural** is the only authoring change. Current CLDR `es` cardinal
= `{one, many, other}`; `many` fires only on large/compact numbers (e.g. millions), but every
`es` ICU plural message must still carry a `many {?}` arm or `intl-messageformat` falls back.
Flag it to the reviewer and extend the parity/lint check. (th/id/zh/ja/ko are **single-form** ?
trivial; da is 2-form like en.)
44 2. **Fonts split the batch.** es/da/id need **no new font** (Latin). th needs one
**small** OFL font. **zh-Hans/ja/ko need large OFL CJK fonts** ? all OFL-1.1 (bundleable under
ED2), all on `@fontsource-variable`, but multi-MB each.
45 3. **CJK + the offline guard collide.** `@fontsource` serves CJK as ~100+ `unicode-
range` `woff2` chunks, fetched on demand ? efficient *online*, but the SPA's **air-
gapped/offline guarantee** (`hygiene.test.ts` "offline / self-contained") means we must ship
**all** chunks so any glyph renders with no network ? **tens of MB** added with 3 CJK languages.
Mitigations in §4.
46
47 ## 4. Design / architecture
48 **Reuse map (what's reused vs new):**
49 - *Reused unchanged: the negotiation contract (persisted pref ? `?lang=` ? navigator ?
`en`), the ICU catalog format, the `@fontsource` `unicode-range`-scoped self-host pattern
(`main.tsx:7-8`), the key-parity + no-bare-strings + offline CI (`hygiene.test.ts`), the
draft?review pipeline, the `` sync (`i18n/index.ts:59-65`).
50 - *New: 7 `common.json` catalogs; `SUPPORTED_LOCALES` entries (`i18n/index.ts:18`); **4
new Noto font imports; an **es 3-form plural** allowance in the catalog + tooling; a **CJK
lazy-load + offline-subset strategy (Phase C); Thai line-break CSS safety net.
51 - **? Catalog-home (resolved by FRIENDLY B3):** catalogs now live at repo-root
**/locales/{en,kn,hi,es,da,id}/common.json** ? the shared top-level dir from the I18N_PLAN
"D-CATALOG-HOME" diagram. FRIENDLY **B3** hoisted them out of `web/src/i18n/locales/` (the SPA
re-points its imports there; the `site/` shim consumes a `copy:locales` mirror). These es/da/id
catalogs moved with the hoist, as anticipated.
52
53 **Negotiation tests (per FRIENDLY §6.5):** every new locale gets an N1/N2-style test ? a

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browser set to that language ****auto-renders**** it (e.g. `navigator.languages=["ja-JP"]` ? `ja``), region subtags normalize (``zh-Hans-CN` ? `zh-Hans``), and a missing key falls back to ``en``. ****Auto-selection is the promise; it's proven by a test, never assumed.****

54

55 ****Phasing (least new code/assets first):****

56 - ****Phase A ? Latin trio (`es`, `da`, `id`): catalog-only, zero new assets.**** Highest ROI, proves the pipeline scales beyond Indic with no code. Only ``es`` needs the 3-form-plural touch.

57 - ****Phase B ? Thai (`th`): one small OFL font.**** Reuses the kn/hi font pattern exactly; add the no-space word-wrap CSS safety net. Self-contained, low-risk.

58 - ****Phase C ? CJK (`zh-Hans`, `ja`, `ko`): the heavy slice.**** Needs the zh-Hans/Hant decision, 3 large OFL fonts, ****per-locale lazy font-load**** (so non-CJK users never download CJK), and an ****offline-bundle subset strategy**** (ship-all-chunks vs a custom subset build step vs online-lazy-only). One PR per CJK language.

59

60 **## 5. Privacy / scope note**

61 i18n is ****presentation-only**** ? no new personal-data collection, no biometrics, minors-excluded posture unchanged (same as FRIENDLY \$5). Gemini-draft translation is inference, not training, with no user data in the loop. CJK/Thai font bundling is OFL media, not code/weights.

62

63 **## 6. Honest limits ? what this does NOT do**

64 - A green CI / DONE proves ****key-parity + render + no-bare-strings + auto-negotiation****, ****not**** translation **quality** (native-review-gated, ED3) or correct font rendering (owner-eyeball-gated, like kn/hi).

65 - ****`zh-Hant` (Traditional) is out of v1**** (separate catalog + font); only ``zh-Hans`` ships first.

66 - ****No RTL**** ? this batch is all LTR.

67 - The ****CJK offline bundle is large**** (tens of MB) unless a subset build is adopted ? a real size trade-off, not free.

68 - ``site/`` marketing localization for these 7 ****depends on FRIENDLY B3**** landing first (the shared shim); until then this is SPA-only.

69

70 **## 7. Deliverables & progress tracker ? **source of truth****

71 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. ****One small PR per row****, branched from ``origin/master``, no stacking.

72

ID	Deliverable	Phase	Depends on	Gated?	Status	PR
E0	This proposal doc + inbound pointers (README/NEXT_STEPS/memory)	?	?	?	**Owner approval to start**	?
EA1	**Spanish (`es`)** catalog (Gemini-draft) + ICU **3-form `{one,many,other}`** handling + extend key-parity/no-bare-strings tooling for the <code>`many`</code> arm + negotiation test	A	E0	No (after approval)	?	[#67](https://github.com/avidullu/khelsutra/pull/67)
EA2	**Danish (`da`)** catalog + negotiation test	A	E0	No	?	[#67](https://github.com/avidullu/khelsutra/pull/67)
EA3	**Indonesian (`id`)** catalog + negotiation test	A	E0	No	?	[#67](https://github.com/avidullu/khelsutra/pull/67)
EB1	**Noto Sans Thai** bundle (OFL, <code>`@fontsource-variable`</code>) + Thai no-space word-wrap CSS safety net	B	E0	No (ED2 carve-out applies)	?	?
EB2	**Thai (`th`)** catalog ? render + negotiation test	B	EB1	No	?	?
EC0	**Decision + design:** <code>`zh-Hans`</code> vs <code>`zh-Hant`</code> (ED6) + CJK **lazy per-locale font-load** + **offline-bundle subset** strategy (ship-all-chunks vs custom subset vs online-lazy)	C	E0	**Owner (Q2/Q3)**	?	?
EC1	**Noto Sans SC** + **`zh-Hans`** catalog + lazy-load + negotiation test	C	EC0	No (after EC0)	?	?
EC2	**Noto Sans JP** + **`ja`** catalog + lazy-load + negotiation test	C	EC0	No	?	?
EC3	**Noto Sans KR** + **`ko`** catalog + lazy-load + negotiation test	C	EC0	No	?	?
EX1	**Native-speaker review** of each AI-drafted catalog (one tracked row; per-language sign-off)	?	the catalog rows	**Owner (reviewer sourcing)**	?	?
EX2	**Owner-machine font eyeball** per new script (th, zh, ja, ko) ? cloud CI can't render fonts	?	EB*/EC*	**Owner (owner machine)**	?	?
EX3	**`site/` marketing parity** for the 7 (rides FRIENDLY **B3** shared <code>`t()`</code> shim)	?	FRIENDLY B3	Blocked on B3	?	?

```

88
89     ## 8. Open questions ? owner
90     1. **Approve starting this line** (in parallel with the FRIENDLY UX/A-items), and the
**A ? B ? C phasing**? *(E0 gate.)*
91     2. **Chinese:** ship **`zh-Hans` only** for v1 (recommended), or **both Hans + Hant**?
*(EC0; Hant = a whole extra catalog + Noto Sans TC.)*
92     3. **CJK offline strategy** *(EC0; the real size trade-off):* (a) ship **all** font
chunks ? simplest, **+tens of MB** to the air-gapped bundle; (b) a **custom subset build**
restricted to glyphs the reviewed catalogs use ? KB-scale but adds a build step that re-runs on
catalog change; (c) **online lazy-load only** ? small bundle but **breaks the offline guarantee
for CJK**. *Recommend (b) if offline-for-CJK matters, else (a).*
93     4. **Native-reviewer sourcing** for 7 languages ? who/how/budget? (Vendor like the Roora
labeling pilot? community? paid freelancers?) This is the launch-gate cost.
94     5. **Priority order** ? which markets first? (Default A-then-B-then-C is by *engineering
cost*; if a market is urgent, e.g. `id` or `ja`, it can jump within its phase.)
95     6. **Does `site/` localize for these too** for v1, or SPA-only until FRIENDLY B3 lands?
*(EX3.)*
96
97     ## 9. Definition of done
98     - [ ] All 7 catalogs (`es`, `da`, `id`, `th`, `zh-Hans`, `ja`, `ko`) live +
**rendering** on the `web/` SPA; switcher + persistence + `en` fallback; **no raw keys**.
99     - [ ] **Auto-negotiation green** ? a browser set to each language auto-renders it
(N1/N2-style tests per locale); region subtags normalize; missing-key ? `en`.
100    - [ ] **CI key-parity + no-bare-strings green** (incl. the `es` 3-form plural) + the
**offline/self-contained guard** still green after the CJK fonts land.
101    - [ ] **Fonts render correctly** ? owner eyeball per new script (th, zh-Hans, ja, ko).
102    - [ ] **Native-speaker review** complete + signed off per language (EX1).
103    - [ ] CJK **offline/bundle-size** strategy implemented per the EC0 decision; documented
bundle-size impact.
104    - [ ] *(if in scope)* `site/` marketing renders the 7 via the shared shim (EX3, after
FRIENDLY B3).
105    - [ ] $7 all rows ? (or explicitly deferred) ? flip **Status ? DONE** + archive.
106
107    ## 10. References
108    **Internal `[verified]`:** `web/src/i18n/{index.ts,hygiene.test.ts}`, `web/src/main.tsx`
(font-load pattern), repo-root `/locales/{en,kn,hi,es,da,id}/common.json` (hoisted by FRIENDLY
B3 ? D-CATALOG-HOME); ['FRIENDLY_AND_MULTILINGUAL.md'](FRIENDLY_AND_MULTILINGUAL.md), engine
`docs/I18N_PLAN.md`, `docs/UI_PROPOSAL.md`.
109    **External `[researched]`:** Unicode CLDR v48 plural-rules chart
(https://www.unicode.org/cldr/charts/48/supplemental/language\_plural\_rules.html) + plural-rules
spec; W3C BCP-47 / Chinese script subtags (`zh-Hans`/`zh-Hant`); SIL Open Font License 1.1 +
Noto CJK LICENSE (github.com/notofonts/noto-cjk); @fontsource-variable/noto-
sans-{thai,sc,jp,kr}; W3C Thai line-breaking (`thai-gap`); react-i18next + i18next-icu.
110
111    ---
112
113    ### Changelog
114    - `2026-06-21` ? **Phase A shipped ?
[#67](https://github.com/avidullu/khelsutra/pull/67)** (E0/EA1/EA2/EA3 ?). es/da/id catalogs
(AI-draft, native review pending) + `lang.*` autonyms + `SUPPORTED_LOCALES`/`LanguageSwitcher`
wiring; **es 3-form {one,many,other}** plural; render + auto-negotiation + plural tests. Added
a **placeholder-parity** hygiene guard (every locale keeps en's` {placeholder}` set) which
caught + fixed a real kn/hi A1-drift (stale `{videoId}` in `mode.loading`/`mode.error`). 168 web
tests green; browser-verified es+da render with all 6 languages in the switcher. Status `DRAFT ?
IN PROGRESS`. Phases B/C still owner-gated.
115    - `2026-06-21` ? created. Owner ask: add
Thai/Indonesian/Danish/Spanish/Chinese/Japanese/Korean with en/kn/hi-level diligence, as a
parallel line for approval. Grounded in a per-language i18n research pass (CLDR plurals, OFL
Noto fonts, BCP-47, offline-bundle sizing) + the verified working-tree foundation. Phasing A
(Latin) ? B (Thai) ? C (CJK); decisions ED1?ED6; 6 owner questions surfaced (E0 = approval
gate).
116

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1      # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3      > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) .
**Owner:** Avi Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE).
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory
`current-state` + `next-session-ux-localization`.
6      > **Relation to existing docs:** this is the **implementation project** for two engine
design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-
indexer/docs/I18N_PLAN.md` (localization) and the UX half of `~/docs/UI_PROPOSAL.md` (non-tech-
savvy console). Those stay the canonical *design*; this doc carries the *tracked execution*
against today's surfaces.
7      > **Honesty note:** each claim tagged `[verified]` (checked against code this session),
`[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font
question (D-FONT) is owner/counsel-gated.
8
9      ---
10
11     ## 0. TL;DR
12     A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their
camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-
holding**. This project isolates that goal into one tracked workstream-pair: ** (A) ** make the UI
super friendly for non-tech-savvy users, and ** (B) ** localize to Hindi + Kannada. Both surfaces
(`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the
foundation ? shared `locales/{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the
real deliverable; translations follow. **The single most important caveat:** translations are
**human-reviewed** (Gemini drafts are *inference only, never training* ? the standing
guardrail), and Kannada/Hindi rendering is **blocked on the OFL font decision (D-FONT)**. The
cheapest unblocked high-impact first slice is the **SPA i18n scaffold + language switcher (en-
only)** ? it locks the shared catalog/negotiation contract and adds the render-test safety net
the SPA currently lacks.
13
14     ## 1. Problem & goal
15     **Why now.** The public site (kheLSutra.guru) is live, clean, honest, and tested. The
PMF beachhead is **Bangalore badminton academies**; many coaches, parents, and players are far
more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified
? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`,
`KHELSUTRA_PRODUCT_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at
the exact point of first contact.
16
17     **Concrete outcome ("good" looks like):** a Kannada/Hindi-comfortable, phone-first, non-
technical coach/parent can ? without help ? (1) understand what the product does and how to
record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4)
share it on WhatsApp, **entirely in their language**.
18
19     **Read to get current:** this doc's $3?$4; the engine design `I18N_PLAN.md` (canonical
i18n design) + `UI_PROPOSAL.md` $2 (personas, the 6 alpha screens, "teach recording *before*
upload"); `VIDEO_RECORDING_GUIDELINES.md`; this repo's `PER_RALLY_SELECTION.md` (the honest SPA
scope) + `REQUEST_ACCESS_FORM.md`.
20
21     ## 2. Decisions locked (structural ? owner directive 2026-06-21)
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | **Isolate as ONE tracked project** spanning both workstreams (A: UX, B: kn/hi
l10n) | owner, 2026-06-21 | This doc; $7 = source of truth; archived on DoD |
25     | D2 | **Home = `kheLSutra/docs/`** (the implementation repo) | owner-gated tracked-doc
convention | `web/` + `site/` both live here; $7 tracks kheLSutra PRs; engine P2 backend-l10n is
a cross-repo deferred row |
26     | D3 | **Adapt `I18N_PLAN.md` to today's surfaces:** P1 (vanilla `t()` shim) ? `site/`;
P3 (react-i18next) ? `web/`; **drop the dead `backend/api/static` retrofit** | `[verified]` the
doc targets `backend/api/static` (`I18N_PLAN.md:25,30,156,191`) which exists only in the engine
repo, not here | The doc's framework-agnostic *foundation* (catalog format + negotiation + ICU
plurals + CI) survives unchanged; only the wiring re-points |

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27 | D4 | **Engine P2 backend-localization stays owner-gated + cross-repo** (not in this
DoD) | `CLAUDE.md` guardrails | Dynamic engine error/report text localizes later; tracked as a
deferred row |
28 | D5 | **Archive trigger = en+kn+hi live on BOTH surfaces** (+ the non-tech UX
baseline), then run the engine `DOC_STATUS.md §2` archival checklist | owner, 2026-06-21 | See
§9 |
29 | D6 | **D-LOCALES: `en` + `kn` for v1, `hi` next; always-fallback to `en`** | owner,
2026-06-21 | Kannada-first beachhead; `hi` = catalog-only (both 2-form CLDR) |
30 | D7 | **D-FONT: OFL-for-media carve-out ? self-host Noto Sans Kannada/Devanagari**
(counsel may still confirm) | owner, 2026-06-21 | Unblocks B4/B5/A3 + the kn/hi reel watermark;
an explicit exception to the MIT/Apache/BSD code guardrail (fonts = media) |
31 | D8 | **D-SURFACE-FIRST: lead with the `web/` SPA scaffold; SKIP the dead static-UI
retrofit** | owner, 2026-06-21 (via B1 go-ahead) | B1 is greenlit; site recording-education
localized early |
32 | D9 | **D-CATALOG-HOME: co-locate `locales/` + CI in `khelsutra`** (shared, consumed by
SPA + site) | owner, 2026-06-21 (via B1 go-ahead) | No cross-repo coupling |
33
34 *Only **D-BACKEND-DEPTH** (§8.3) remains open ? and it's a deferred, cross-repo engine
concern (B7), not a v1 blocker.*
35
36 ## 3. Foundation ? research / prior art
37 **Current-state facts `[verified]` this session** (adversarially checked against the
working tree):
38 - **`web/` SPA: zero i18n.** `web/package.json` deps are only `react`/`react-dom`; no
`i18next`/`react-intl`/`formatjs`; no `locales/` dir; no `t()`/`useTranslation`/`data-i18n`
anywhere. User-facing English is hardcoded in `*.tsx` (e.g. `QueryBar.tsx:46-47,56,72,76,78`;
`App.tsx:115,211,224`) ? ~50 literals + ~16 derived labels.
39 - **`site/` marketing: vanilla, zero i18n.** Three `` files
(`public/{index,capture,own-model}.html`) with inline CSS, no framework; the only scripts are a
CF analytics beacon (`index.html:595`) + one inline form/reel IIFE (`index.html:597-633`).
`server.js` hardcodes English responses; no locale handling.
40 - **Recording-education is *not* greenfield.** The homepage "Record it right" gist
(`index.html:324-364`) + the `/capture` checklist (`capture.html`, routed `server.js:51`,
guarded `assets.test.js:94`) already ship the user-facing best-practices surface. *Unshipped:*
the empirical-threshold framing and the programmatic pre-flight validator (a code change, not a
page).
41 - **Persona is design intent, not built.** kn/hi + zero-knobs are **proposed/owner-
gated**; the app is English-only today (`KHELSUTRA_PRODUCT_OVERVIEW.md:61`). It's
Kannada-**first**, Hindi-**next** ? not Kannada-only.
42
43 **Localization prior art `[researched]`:
44 - **CLDR plural categories:** Hindi = `{one, other}`; Kannada = `{one, other}`. Both are
2-form (same shape as English) ? once ICU MessageFormat is in the catalog, adding `hi`/`kn`
plurals is **catalog-only**, no code. (ICU still required because English `count>1?s:'` hand-
rolling ? present today in the SPA ? doesn't generalize.)
45 - **react-i18next** (web/): `i18next` + `react-i18next` + `i18next-browser-
languagedetector` + an ICU/plural post-processor; lazy-load per-locale JSON; persist via
`localStorage`.
46 - **Shared catalog + tiny shim** (site/): the SAME i18next-flavored JSON consumed by a
~30-line `t(key, vars)` shim that walks `data-i18n` attributes ? no build step, no framework,
mirrors i18next interpolation/plural rules for the 2-form languages.
47 - **Noto Sans Kannada / Noto Sans Devanagari** are **OFL-1.1**; OFL permits commercial
bundling/redistribution (fonts = media, not code/weights). Self-host (e.g. `@fontsource`) with
`font-display: swap` + subsetting. **Gated on D-FONT.**
48 - **Non-tech-savvy / HCI4D patterns:** icon **+ text** (never icon-only), non-color
selection indicators, large touch targets, mobile-first, WhatsApp as the share channel, a
prominent persistent language switcher.
49
50 ## 4. Design / architecture
51 **Two surfaces, one shared source of truth.**
52
53 ...
54 locales/ # shared catalogs ? co-located in khelsutra (D-CATALOG-
HOME, see §8)
55 en/ common.json # source of truth ? devs author keys here

```



```

56      kn/  common.json          # Kannada  (Gemini-draft ? human-review)
57      hi/  common.json          # Hindi    (next ? the extensibility proof)
58
59      web/  ? react-i18next      # I18N_PLAN P3: provider in main.tsx; useTranslation in
components
60      site/ ? t(key,vars) shim   # I18N_PLAN P1 retargeted: data-i18n attrs over the
static HTML
61      ```
62
63      - **Negotiation contract (one order, both surfaces):** `persisted pref (localStorage) ?
switcher / ?lang=<code> ? navigator/Accept-Language ? "en"`. Supported locales in a single
module; anything else falls back to `en`; **a missing key degrades to English, never to a raw
key.**
64      - **Keys, never prose** ? `t("compile.button")`, with ICU for interpolation/plurals:
`"${count, plural, one {# rally} other {# rallies}}". This replaces the SPA's current hand-
rolled `Segment${count>1?'s':''}`.
65      - **Translation pipeline (guardrail-clean):** dev authors `en` keys ?
`scripts/i18n_draft.py`-style Gemini **draft** (inference only, never training; flags every
value for review) ? **owner/native-speaker review** ? CI key-parity ? merge. Gemini never
touches `en`, never invents keys.
66      - **CI enforcement (keeps it "by design"):** (1) **key-parity** ? every locale has
exactly the `en` key set (drop a stub `ta.json` ? CI lists every gap = the extensibility proof);
(2) **no-bare-strings** ? flags user-facing literals not routed through `t()` (curated allowlist
to start).
67      - **Fonts:** bundle Noto Sans Kannada + Devanagari **iff D-FONT = OFL carve-out**; this
also unblocks a Kannada/Hindi **reel watermark** (engine `WatermarkConfig.font_path` already
takes a path ? config, not code).
68
69      **Non-tech-savvy UX levers** (ranked impact × effort, grounded in personas; surface-
tagged):
70
71      | Lever | Surface | Impact | Effort | Note |
72      |---|---|---|---|---|
73      | Plain-language copy ? strip engine jargon, keep honest caveats | both | high | small |
banners + footer carry backend vocabulary today |
74      | Localized **WhatsApp share** for the downloaded reel | both | high | small | WhatsApp
is *the* channel for this audience |
75      | **Language switcher** persisted via a shared localStorage key | both | high | small |
site promises kn/hi but ships no switcher |
76      | Localize **recording-education** (capture + gist) kn?hi | site | high | medium | gates
input quality; the highest-leverage kn content |
77      | Mobile-first touch targets; tooltip-text ? visible | web | high | medium | tiny
targets + desktop layout + invisible tooltips on phones |
78      | Icon **+ text** + non-color selection indicator | web | medium | small | never icon-
only |
79      | Friendly engine-error mapping over raw FastAPI `detail` | web | medium | medium |
`CompileResult` passes engine text untranslated |
80      | Plain-language framing for the Request-Access GPU/storage Qs | site | medium | small |
"GPU"/"S3/GCS" are intimidating |
81      | Background-processing reassurance + status as icon chips | web | medium | small |
honest expectations on slow uplinks |
82      | Fix English-source recording gaps (a diagram) before translating | site | medium |
medium | the diagram is reusable across all 3 languages |
83      | Tap-only filter controls alongside the English query bar | web | medium | large | the
`parseQuery()` parser is English-grammar only |
84      | Keep the honest *today-vs-roadmap* split through translation | both | medium | small |
never soften "experimental"/"not available yet" |
85
86      **Reuse map (what's new vs reused):** **Reused** ? the i18next-flavored JSON + negotiation
contract + ICU plurals + key-parity/no-bare-strings CI + draft?review pipeline (all from
`I18N_PLAN.md`); the shipped recording pages; the SPA's `lib/format.ts`/`lib/query.ts`;
`site/src/assets.test.js` + the SPA's vitest harness. **New** ? `locales/` catalogs; the SPA
react-i18next provider + switcher; the site `t()` shim + `data-i18n` wiring; Noto font bundle
(gated); a jsdom render-test env for the SPA.
87

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88     ## 5. Privacy / scope note
89     i18n touches **presentation only** ? no new personal-data collection, no biometrics,
minors-excluded posture unchanged. WhatsApp share opens the user's own client with a pre-filled
message; it sends nothing server-side. Gemini-draft translation is inference, not training (no
user data in the loop).
90
91     ## 6. Honest limits ? what this does NOT do
92     - A green CI / DONE status proves **key-parity + render + no-bare-strings**, **not**
translation *quality* ? that's human-gated and owner-paced.
93     - Kannada/Hindi will **not render** until the Noto fonts land (D-FONT); until then kn/hi
catalogs exist but display tofu/fallback.
94     - The SPA is **job-tethered**; localizing **dynamic engine errors/report text** needs
engine P2 (owner-gated, cross-repo, D-BACKEND-DEPTH) ? out of this DoD.
95     - The query bar stays **English-grammar parsing**; kn/hi users rely on tap-only controls
(a later, larger slice), not free-text Kannada queries.
96     - **Mobile-*perfect** layout is an alpha non-goal (usable-on-phone is in scope; pixel-
perfect is not).
97
98     ## 6.5 Testing strategy ? automatic language negotiation (rigorous, owner-required)
99     **The bar (owner, #41):** a user whose **computer/browser language is Kannada or Hindi
must get the respective version automatically ? that is the core promise and it is **proven by
tests at every layer, never assumed. **Auto-selection IS the negotiation order (persisted pref ?
`?lang=` ? browser/`Accept-Language` ? `en`); each step gets a test.
100
101     **Negotiation matrix ? every row is an automated test:**
102
103     | # | Signal under test | Setup | Expected | Proven in |
104     |---|---|---|---|---|
105     | N1 | **Browser/OS = Kannada** | no persisted pref, no `?lang`,
`navigator.languages=["kn-IN", ?]` | resolves to **kn** ? Kannada content (once the kn catalog
ships) | **B1** detection . **B5** content |
106     | N2 | **Browser/OS = Hindi** | `navigator.languages=["hi-IN", ?]` | resolves to **hi**
? Hindi content | **B6** (hi added to the supported set + catalog) |
107     | N3 | **`?lang=` override** | `?lang=kn`, browser = en | kn | B1 |
108     | N4 | **Persisted pref wins** | `localStorage=kn`, browser = en, no `?lang` | kn (a
returning user keeps their choice) | B1 |
109     | N5 | **Region subtag normalizes** | `navigator.language="kn-IN"` | base **kn** (not a
"kn-IN" miss) | B1 |
110     | N6 | **Unsupported language** | `navigator="ta-IN"` (Tamil, not yet supported) |
**en** fallback ? never a locale we can't serve | B1 |
111     | N7 | **Missing key in active locale** | kn active, a key absent from `kn/common.json`
| **en** fallback for that key (no raw key shown) | B1/B5 |
112     | N8 | **Static site (vanilla shim)** | `Accept-Language: kn` / `?lang=kn` on a `site/`
page | kn-rendered page (same order as the SPA) | B3 |
113     | N9 | **Backend (engine P2)** | `Accept-Language: kn` to the API | kn customer/report
text | B7 (owner-gated) |
114
115     **Levels & tooling:**
116     - **SPA (`web/`):** jsdom tests stub `navigator.language`/`navigator.languages`,
`localStorage`, and the `?lang` query ? assert `i18n.language`/`resolvedLanguage` + rendered
text per the matrix. *(B1 lands N1/N3?N7 at the **detection** level now ? `kn` is supported; the
kn/hi **content** assertions ride the catalogs in B5/B6.)*
117     - **Static site (B3):** the shared `t()` shim's negotiation is unit-tested the same way;
plus a server test that `Accept-Language: kn`/`?lang=kn` returns the kn-rendered page.
118     - **CI key-parity:** guarantees every locale has every key, so a correctly-negotiated
locale can never fall through to a raw key.
119     - **End-to-end (manual, owner machine ? per I18N_PLAN $6):** set the real OS/browser to
Kannada, then Hindi ? load both surfaces ? confirm the right language renders **and the Noto
fonts display** (the cloud CI container can't render fonts; the owner eyeball is the final
gate).
120     - **? Query parsing / semantic understanding (forward-looking, owner directive
2026-06-21) ? i18n testing is MANDATORY there. **The query bar is **English-grammar-only** today
(a documented limit, tracker A9): the user types English queries (`over 10s`, `top 15 longest`)
and a localized **sample-query prompt** shows them what to type in *every* locale (kn/hi
included). The **moment** the parser gains **semantic understanding** or accepts **Kannada/Hindi

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query input** (e.g. typing a Kannada/Hindi phrase, or a conversational/NL builder ? I18N_PLAN
P4), i18n tests become a **merge gate**: a kn/hiquery (typed or spoken) must parse to the
**correct selection**, the localized **parse-feedback** ("parsed-pill") must render in the
active locale, and the unsupported/empty-result messages must localize. **No query-NL /
semantic-parsing work merges without those tests.**
121
122     A locale that's reachable by the switcher but **not auto-selected from the browser
language is a bug, not a "later". ** This matrix is a DoD gate ($9).
123
124     ## 7. Deliverables & progress tracker    ? **source of truth**
125     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking.
126
127     | ID | Deliverable | Workstream | Depends on | Gated? | Status | PR |
128     |----|-----|-----|-----|-----|-----|----|
129     | P0 | This tracked project doc + inbound pointers | ? | ? | No | ? |
[#41](https://github.com/avidullu/khelsutra/pull/41) · engine
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275) |
130     | B1 | **SPA i18n scaffold + language switcher, en-only** (react-i18next + detector +
`en` catalog + switcher + persistence + **jsdom render tests** + **browser-language negotiation
tests** $6.5 N1/N3?N7) | B | P0 | No | ? | [#42](https://github.com/avidullu/khelsutra/pull/42)
|
131     | B2 | Extract SPA strings ? `en` catalog (~50 literals + ~16 derived; ICU plural keys);
add **key-parity + no-bare-strings CI** | B | B1 | No | ? |
[#45](https://github.com/avidullu/khelsutra/pull/45) |
132     | A1 | Plain-language copy pass on the `en` catalog (de-jargon; icon+text; non-color
selection) | A | B2 | No | ? | [#60](https://github.com/avidullu/khelsutra/pull/60) |
133     | B3 | **Shared `t()`/`data-i18n` shim + persistent switcher for `site/`** ? hoist
catalogs to repo-root `~/locales` (D-CATALOG-HOME), re-point the SPA, site copy-step + drift-
guard, `server.js` + shim negotiation ($6.5 **N8**); **en-only marketing content** (kn/hi rides
A3) | B | B2 | D-CATALOG-HOME ? | ? | [#70](https://github.com/avidullu/khelsutra/pull/70) |
134     | A2 | Localized **WhatsApp share** for the reel ? SPA (CompileResult) + site (post-
signup reveal via a shim `window.KhelSutra.t` bridge); `wa.me`, client-only, `rel=noopener` | A
| B1/B3 | No | ? | [#82](https://github.com/avidullu/khelsutra/pull/82) |
135     | A6 | Plain-language framing for Request-Access GPU/storage questions (de-jargoned +
GPU reassurance; wired via `data-i18n`, en ? form kn/hi is a later increment) | A | B3 | No | ?
| [#75](https://github.com/avidullu/khelsutra/pull/75) |
136     | A7 | **Camera-placement diagram** on the capture page (accessible inline SVG: elevated
camera behind the baseline, ~30° tilt, full-court field-of-view) with a localized en/kn/hi
legend | A | ? | No | ? | [#76](https://github.com/avidullu/khelsutra/pull/76) |
137     | B4 | **Noto Sans Kannada + Devanagari** bundle (OFL carve-out) ? **SPA done; reel
watermark `font_path` = engine follow-up* | B | ? | D-FONT ? | ? |
[#46](https://github.com/avidullu/khelsutra/pull/46) |
138     | B5 | **Kannada (`kn`) catalog** ? render ? **SPA done (AI-draft, native review
pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? |
[#47](https://github.com/avidullu/khelsutra/pull/47) |
139     | A3 | Localize **recording-education** (capture + homepage gist) + the `site.*`
marketing copy to **kn + hi** (Noto on the site; AI-draft, adversarially verified, native review
pending) | A | B3/B5 | D-LOCALES ? + D-FONT ? (site: Google-Fonts Noto) | ? |
[#74](https://github.com/avidullu/khelsutra/pull/74) |
140     | A4 | Mobile-first touch targets + tooltip-text ? visible (SPA) | A | B1 | No | ? |
[#61](https://github.com/avidullu/khelsutra/pull/61) |
141     | A5 | Friendly engine-error mapping over raw FastAPI `detail` (SPA, client-side map) |
A | B2 | No | ? | [#62](https://github.com/avidullu/khelsutra/pull/62) |
142     | B6 | **Hindi (`hi`) catalog** ? render (the extensibility proof ? just a catalog) ?
**SPA done (AI-draft, native review pending)* | B | B5 | D-LOCALES ? | ? |
[#48](https://github.com/avidullu/khelsutra/pull/48) |
143     | A8 | Background-processing reassurance + verdict/status as icon chips (SPA) | A | B2 |
No | ? | [#63](https://github.com/avidullu/khelsutra/pull/63) |
144     | A9 | Tap-only filter controls alongside the English query bar (SPA) | A | B2 | No | ?
| [#68](https://github.com/avidullu/khelsutra/pull/68) |
145     | B7 | ***(deferred, cross-repo)* Engine **P2 backend customer-facing localization**
(report `customer_*` + rejection reasons) | B | engine | **D-BACKEND-DEPTH, owner-gated** | ? |
? |
146

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147 *** MILESTONE ? SPA fully trilingual (2026-06-21):** the `web/` Studio SPA now speaks
en + kn + hi end to end (B1?B2?B4?B5?B6, all merged). A **Kannada-default browser auto-
renders Kannada**, and the switcher walks kn ? hi ? en ? **proven by a Playwright screencast**
with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames captured per locale + `lang>` tracked). This meets the owner's success criterion **on the SPA surface**. **B3 has since
landed** ? the `site/` marketing surface now carries the shared `t()/`data-i18n` shim, a
persistent en/kn/hi switcher, and matching negotiation (client shim + `server.js`; \$6.5 **N8**
green), with **en-only marketing content** for now (the shim/switcher/negotiation are the proof;
kn/hi *marketing* translation rides **A3**). **A3 has since landed** ? the `site/` recording-
education (homepage gist + capture checklist) and the `site.` marketing copy now render in **kn
+ hi** (Noto bundled on the site), so a Kannada/Hindi browser auto-gets localized recording
instructions. **A6** (plain-language access form,
[#75](https://github.com/avidullu/khelsutra/pull/75)) and **A7** (camera-placement diagram,
[#76](https://github.com/avidullu/khelsutra/pull/76)) have since landed, and the **kn/hi SPA-
console native-review pass** shipped ([#77](https://github.com/avidullu/khelsutra/pull/77)).
Still open for the full \$9 DoD: a **native-speaker review** of the *remaining* AI-drafted
catalogs (es/da/id + the reel-watermark strings; the kn/hi SPA console is done via #77) before a
wide public launch. (**A2** WhatsApp/social share is **deferred out of v1 scope** ? not a DoD
blocker.) Do **not** archive until both surfaces render kn/hi *content*. ? (both now do, pending
the remaining native review).

148

149 ## 8. Open questions ? owner (blocking gates) `[design]`

150 *Status (2026-06-21): 4 of 5 **LOCKED/ADOPTED** by the owner ? see \$2 (D6?D9). Only
D-BACKEND-DEPTH remains open (and it's deferred, not a v1 blocker).*

151 1. **D-FONT ? ? LOCKED ? (a) OFL-for-media carve-out, self-host Noto Sans
Kannada/Devanagari** (counsel may still confirm). An explicit exception to the MIT/Apache/BSD
code guardrail (fonts = media). Unblocks B4/B5/A3 + the kn/hi reel watermark.

152 2. **D-LOCALES ? ? LOCKED ? `en` + `kn` for v1, `hi` next, always-fallback to `en`**
(both 2-form CLDR ? `hi` is catalog-only).

153 3. **D-BACKEND-DEPTH ? ? OPEN (deferred, not a v1 blocker).** Recommend **customer
verdict/report `customer\_` + validation-rejection reasons first; technical/operator text stays
English; phase two** (engine repo, owner-gated ? tracker row B7).

154 4. **D-SURFACE-FIRST ? ? ADOPTED ? lead with the SPA scaffold (B1), localize site
recording-education early, SKIP the dead static-UI retrofit.**

155 5. **D-CATALOG-HOME ? ? ADOPTED & IMPLEMENTED (B3) ? repo-root
`/locales/{en,kn,hi,es,da,id}/common.json` is the single source of truth.** The SPA imports it
directly (re-pointed to `../..../locales`; Vite resolves the workspace root via the repo `.git`
marker, so dev/build/test all work). The dependency-free site consumes it through one tiny build
step ? `npm run copy:locales` mirrors `/locales` ? `site/public/locales/` (auto-runs
`predeploy`), with a **drift-guard test** (`site/src/locales-drift.test.js`) failing CI if the
shipped copy goes stale. (The engine `I18N\_PLAN` assumed the indexer repo; this is the
adaptation.)

156

157 ## 9. Definition of done

158 - [x] **Automatic language negotiation (\$6.5) is green** ? N1/N3?N7 (SPA, jsdom) + N2
(hi) + N8 (static-site shim honors `Accept-Language`/?lang`, `server.test.js`), all in CI.

159 - [] **Manual cross-check (owner machine):** OS/browser set to Kannada and to Hindi
each render the respective language on BOTH surfaces, with Noto fonts displaying. *(Owner
eyeball ? pending; cloud CI can't render font glyphs. Closest proxy: the Playwright screencasts
in `khelsutra-screencast/` ? `khelsutra-site-i18n-desktop.mp4` (site, kn) +
`khelsutra-i18n-screencast.mp4` (SPA, `locale: kn-IN`)).*

160 - [x] `en + kn + hi` catalogs live and **rendering** on BOTH `web/` SPA and `site/`
marketing (switcher + persistence + `en` fallback; no raw keys shown) ? B1?B6 (SPA) + B3/A3
(site), proven by render tests + a real-browser pass.

161 - [] **Noto Sans Kannada + Devanagari render correctly** ? owner eyeball on a real
machine. *(SPA self-hosts OFL Noto (B4); the site loads Noto via Google Fonts (A3). Pending the
owner's device check.)*

162 - [x] CI **key-parity + no-bare-strings** green; a stub locale flags every gap ?
`web/src/i18n/hygiene.test.ts` (+ placeholder-parity, es 3-form) + the site `locales-drift`
guard.

163 - [x] **Recording-education** (capture + homepage gist) available in `kn` + `hi` ? A3
(#74), incl. a localized camera-placement diagram (A7, #76).

164 - [x] **Non-tech UX baseline** ? plain-language copy (A1/A6), mobile tap targets (A4) +
the decluttered compact switcher (#83), icon+text status (A8), WhatsApp share (A2), tap-only
filters (A9), honest today-vs-roadmap preserved through translation.

165 - [] \$7 all rows ? (or explicitly deferred) ? flip **Status ? DONE** + run the engine
`DOC\_STATUS.md \$2` archival checklist ? `git mv` to `docs/archives/`. *(\$7 is all ? except the
owner-gated, cross-repo **B7** ? deferred. Holding the DONE flip on the two human gates above
(native-speaker review of the AI-drafted kn/hi + the owner font eyeball). The prior engineering
follow-up ? **own-model body content** English on kn/hi ? is now **closed**
([#86](https://github.com/avidullu/khelsutra/pull/86)): its full body is localized, so no
engineering gap remains before the two human gates.)*

166

167 **## 10. References**

168 **Internal code anchors** `[verified]`: **web/package.json**,
`web/src/{App.tsx,main.tsx,components/QueryBar.tsx,components/CompileResult.tsx,lib/query.ts}`;
`site/public/{index,capture,own-model}.html`, `site/server.js`, `site/src/assets.test.js`.
This repo's docs: **docs/PER_RALLY_SELECTION.md**, **docs/REQUEST_ACCESS_FORM.md**,
`docs/README.md`, **NEXT_STEPS.md**. **Engine design docs (canonical)**: **badminton-highlight-indexer/docs/{I18N_PLAN.md,UI_PROPOSAL.md,VIDEO_RECORDING_GUIDELINES.md,PMF_MARKET_RESEARCH.md,PMF_F**
IELD_SIGNALS.md,KHELSUTRA_PRODUCT_OVERVIEW.md,DOC_STATUS.md}. **External** **[researched]**:
Unicode CLDR plural rules (kn/hi = one/other); react-i18next + i18next-icu; SIL Open Font
License 1.1; Google Noto Sans Kannada/Devanagari.

169

170 ---

171

172 **### Changelog**

173 - `2026-06-21` ? created. Isolated the UX + Hindi/Kannada work into one tracked project
(owner directive). Grounded in a multi-agent research+verification pass (web/+site/ string maps,
Indic-l10n prior art, 5 adversarial verdicts). Decisions D1?D5 locked; 5 product/licensing
decisions surfaced as open; \$7 tracker seeded with B1 as the recommended first slice.

174 - `2026-06-21` ? owner locked **D6** (en+kn?hi) + **D7** (OFL Noto carve-out) and
adopted **D8** (SPA-first) + **D9** (catalogs in khelsutra); only D-BACKEND-DEPTH remains open.
B1 greenlit ? ? in progress.

175 - `2026-06-21` ? **B1** opened ? [#42](https://github.com/avidullu/khelsutra/pull/42):
SPA i18n scaffold (react-i18next + ICU + detector) + `en` catalog + language switcher + the
SPA's first jsdom render tests. Verified typecheck/79-tests/build green + live in-browser
(switcher persists, en-fallback). Doc/pointer PRs opened: khelsutra
[#41](https://github.com/avidullu/khelsutra/pull/41), engine
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275).

176 - `2026-06-21` ? **\$6.5** Testing strategy added (owner ask, #41): **automatic-negotiation matrix**
(N1?N9) so a Kannada/Hindi **browser** auto-gets the respective
version ? proven at the detection level now (B1), at the content level with the catalogs
(B5/B6), on the static shim (B3), and by a manual font-render eyeball. Wired into the DoD (\$9).
B1 (#42) gains the detection tests (N1/N3?N7).

177 - `2026-06-21` ? **A9** (#68) landed. **Localized tap-only filter controls** (**TapFilters**:
Sort/Length/Count) so a non-English-typing kn/hi/? user can filter without typing the English
query syntax; each tap builds the same **ParsedQuery** the parser would (tap == type, test-
pinned). The English example chips now show only in English. 173 web tests; reviewed via a
2-lens adversarial workflow. **Workstream A**: A1/A4/A5/A8/A9 ?; remaining A3 (recording-edu
kn/hi), A6/A7 (site), B3 (site trilingual). **(Also note: MULTILINGUAL_EXPANSION Phase A ?**
es/da/id ? shipped #67, so the SPA now speaks 6 languages.)

178 - `2026-06-21` ? **A8** (#63) landed. **Background-processing reassurance + status as**
icon+text chips (never icon-only): a new **StatusChip** + shared **Spinner**; IngestPanel processing
shows a working chip + "safe to wait ? runs on the server" line and a ? "Couldn't process" chip
on failure; CompileResult gains a "Stitching your reel?" building card + ??? status icons on the
terminal card (now `aria-live`). 141 web tests green; reviewed via a 2-lens adversarial workflow
(folded its aria-live + honest-copy findings). **Workstream A** now: A1/A4/A5/A8 ?; remaining A9
(tap-only filters), A3 (recording-education kn/hi).

179 - `2026-06-21` ? **A5** (#62) landed. **Friendly engine-error mapping unified across ALL**
SPA surfaces (ingest + build + rally-load) via a single **lib/errorText.ts** helper + a shared
errors. catalog namespace: an **engine** failure shows the engine's own detail verbatim,
everything else a localized line (**engineGeneric** last resort). No more raw **ApiError**/**HTTP**
500 text; **build.failed**/**rallies.load_failed** logs gain **code**+**detail**. 136 web tests green;
reviewed via a 2-lens adversarial workflow (folded its load-path consolidation nit).

180 - `2026-06-21` ? **A1** (#60) + **A4** (#61) landed. **A1** = the plain-language **en** copy pass
(de-jargon). **A4** = the non-tech / phone-first pass: ?44px tap targets across the SPA (Tailwind
`min-h-11`/`h-11`), enlarged parsed-pill + sample chips, and **hover-only** **title**=` text surfaced
as visible DOM text` (the QueryBar "not available yet" reason + per-card shot-count confidence)
? a phone never shows **title**=`. 128 web tests green; reviewed via a 3-lens adversarial workflow.

****Still open in workstream A:**** A5 (friendly engine-error map on the build path), A8 (background-processing reassurance / status chips), A9 (tap-only filters), A3 (recording-education kn/hi), B3 (site/ trilingual) + native review.

181 - `2026-06-21` ? ****?** SPA MILESTONE: fully trilingual (owner directive "take it to completion + screencast"). ****** B2 (#45, full string extraction + key-parity/no-bare-strings CI) ? B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ? the extensibility proof) all merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6 render + key-parity.

****Playwright screencast**** (`locale: kn-IN`): a Kannada-default browser auto-renders Kannada ? switch hi ? switch en, each with `` tracked (`khelsutra-i18n-screencast.mp4` + per-locale frames). kn/hi are ****AI-drafted ? native review pending****. ****Remaining for full \$9 DoD:**** B3 (the `site/` marketing surface), A3 + the UX A-items, native-speaker translation review.

182 - `2026-06-21` ? ****B3 ([#70](https://github.com/avidullu/khelsutra/pull/70))** landed ? the marketing `site/` is now i18n-wired, en-only content. ****Implements **D-CATALOG-HOME**:** catalogs hoisted to repo-root ****/locales**** (all 6 ? en/kn/hi + the merged es/da/id) as the single source of truth; the SPA re-points its imports there (Vite resolves the `.git` workspace root ? typecheck/168 web tests/build green). The dependency-free site gains its one build step ? `npm run copy:locales` mirrors `/locales` ? `site/public/locales/` (`predeploy` hook) + a ****drift-guard test****. New ****site/public/i18n.js**** ? a ~110-LOC vanilla `t(key,vars)` + `data-i18n` walker with ICU-lite interpolation/plurals via `Intl.PluralRules` (mirrors i18next for the 2-form site languages); a ****persistent en/kn/hi switcher**** on `index/` capture `own-model`; and ****server.js` Accept-Language negotiation\*\*** matching the SPA EXACTLY (persisted cookie ? `?lang=` ? `Accept-Language`/navigator ? en; missing key ? en, never a raw key). Tests: shim unit test + drift guard + extended `assets/` `server` tests + the **\*\*\$6.5 N8\*\*** negotiation suite ? **\*\*62 site tests green\*\***; real-browser smoke (switch ? persist ? reload). kn/hi **\*\*marketing\* translation is the next effort (\*\*A3\*\*)**. ? **\*\*Concurrency note:\*\*** built in an isolated `git worktree` because a sibling ****A9**** session held uncommitted catalog WIP in the shared checkout ? never clobbered.

183 - `2026-06-22` ? ****A3 ([#74](https://github.com/avidullu/khelsutra/pull/74))** landed ? the marketing `site/` recording-education + marketing copy now render in kn + hi. ****Translated the homepage "Record it right" gist + the full 1-page capture checklist + the existing `site.` copy to **Kannada + Hindi**** (~40 new `site.record.`/`site.capture.` keys; en-authored, kn/hi translated, es/da/id en-seeded ? key-parity green). The shim gains a ****data-i18n-html**** form so inline `**` emphasis survives translation; ****Noto Sans Kannada + Devanagari**** bundled via Google Fonts on all 3 pages (+ `--font` fallback) so kn/hi render without tofu. Translations ****AI-drafted ? adversarially verified**** by a 6-agent workflow (meaning/fluency/integrity x kn/hi; 0 issues in the A3 keys) and a 4-lens code review (folded its a11y findings ? switcher group `aria-label` now localized via `lang.label`, default `aria-pressed` for the no-JS baseline ? and 3 test-coverage findings incl. a byte-for-byte HTML?en-catalog drift test). ****Web 173 + site 68 green****; real-browser check (kn capture page renders Kannada, `**` preserved, no console errors). kn/hi remain ****native-review-pending**** (font glyphs = owner eyeball gate). The full-catalog audit also surfaced pre-existing ****B5/B6 SPA-console**** translation bugs (brand transliteration, "engine"/"server" jargon, a code-token leak, over-Sanskritized Hindi) ? flagged as a separate follow-up.****

184 - `2026-06-22` ? ****A6 ([#75](https://github.com/avidullu/khelsutra/pull/75))** landed ? plain-language Request-Access GPU/storage questions. ****De-jargoned the intimidating form copy ("GPU machine that could process video locally" ? "a powerful computer that could process videos on its own"; dropped the bare "(S3/GCS)") and added a **GPU reassurance** ("No worries if not ? we can run it in the cloud for you"), so a non-technical coach/parent isn't scared off. Wired via the shim (`site.join.`), `data-i18n` spans that preserve the radio/checkbox ``s), translation-ready (en for now; the form's kn/hi is a later increment). The ****backend enum contract is unchanged**** (field names + every `has\_gpu`/`storage` value preserved ? server-validated). Tests: a de-jargon + reassurance + contract-intact assertion, and the A3 HTML?en-catalog drift test auto-covers the new keys. Site 69 + web 173 green.**

185 - `2026-06-22` ? ****A7 ([#76](https://github.com/avidullu/khelsutra/pull/76))** landed ? camera-placement diagram on the capture page. ****Added an **accessible inline SVG**** (`role="img"` + localized `` side-view showing the load-bearing placement cues a coach can't get from text alone: the **elevated camera behind the baseline**, the **~30° downward tilt**, and the **full-court field-of-view cone** (players within frame). A numbered ??? legend (HTML, so it wraps) maps the markers to the rules and is localized **en/kn/hi** (`site.capture.diagram.`, reusing the A3 capture terms) ? consistent with the now-trilingual capture page. The text "Place the camera" step remains (the diagram complements it). Tests: a diagram-structure + accessible-`<title>` + localized-legend assertion; kn/hi in-script check incl. `diagram.camera`; the A3 drift test auto-covers the new keys; site 70 + web 173 green. **Workstream A is now A1/A3/A4/A5/A6/A7/A8/A9 ?** (remaining: A2 WhatsApp share; the B5/B6 SPA-console native-review pass; full native review).</p>
</div>

186 - `2026-06-22` ? ****A2** ([#82](https://github.com/avidullu/khelsutra/pull/82)) landed ? localized WhatsApp share (the last open \$7 row). ****SPA** `CompileResult` done-state + the site's post-signup reveal each offer a localized "Share on WhatsApp" CTA that opens `wa.me` with a pre-filled message (client-only; `rel=noopener`). SPA `result.share`/`result.shareText` translated in all 6 locales; the site reveal (injected after sign-up, outside the static `data-i18n` walk) localizes via a new ****window.KhelSutra.t` shim bridge** (`site.reveal.*`, en/kn/hi real + es/da/id seeded). Verified in-browser: the kn reveal renders "? WhatsApp ????? ??????" + a Kannada message. Web 215 + site 72 green. ****\$7** is now all ? except the owner-gated, cross-repo B7** (engine backend l10n ? deferred). Remaining for the DoD: native-speaker review of the AI-drafted kn/hi + the owner's font-glyph eyeball.

187 - `2026-06-22` ? ****UI/UX** header pass ([#83](https://github.com/avidullu/khelsutra/pull/83)). ****A 3-lens design critique + a deep Playwright layout test** (all pages x {1280, 390} x {en, hi/kn}) confirmed the home header ****overflowed on mobile** ? the 3 always-visible language pills pushed the "Request access" CTA off-screen (worse in kn). Replaced the pills with a ****compact `<select>` switcher** on all three pages (shows the current language, can't overflow, one tap target, accessible, consistent), added wrap/`overflow-x` safety, and localized the ****home hero headline** (`site.hero.title`, the last big English string on the kn/hi home page) via `data-i18n-html`. Shim now reflects the locale in `select[data-langsw]` + listens for `change`. ****Zero horizontal overflow on all 12 page/viewport/locale combos**; web 215 + site 72 green. Screenshots (desktop kn journey + mobile header) saved to `khelsutra-screenshot/`.

188 - `2026-06-22` ? ****?** Multilingual chapter ? effectively complete (pending human gates). ****\$7** is ****all ? except B7** (assessed: significant cross-repo ****engine** backend-i18n in Python with no existing infra + owner-gated ****D-BACKEND-DEPTH** ? kept deferred). Both surfaces are trilingual end-to-end (SPA en/kn/hi + es/da/id; site en/kn/hi), non-tech-friendly, and the ****sister repo `rally-annotator`** is also localized (kn/hi/es/da/id/te) ? so the coach's journey ****annotator ? site ? Studio** is consistent in their language. ****What gates archival:** (1) a native-speaker review of the AI-drafted kn/hi catalogs (site + SPA), (2) the owner's font-glyph eyeball on a real Kannada/Hindi device, (3) a small remaining slice ? `own-model` body content is still English on kn/hi. Do not `git mv` to `docs/archives/` until (1)+(2) clear.

189 - `2026-06-22` ? ****own-model** body content localized to kn + hi ([#86](https://github.com/avidullu/khelsutra/pull/86)) ? closes engineering follow-up (3) above. ****The developer/power-user `own-model.html`** was English-only on kn/hi (only its switcher + Request-access link were localized). Wired its ****full body** ? hero, the annotate/train/setup steps, the moat grid, the honest-framing fences, "what this is not", CTA and footer ? through the shim (****52 new `site.ownModel.` keys**; `data-i18n` + `data-i18n-html` for inline `**<b>/<i>/<a>/<code>/<span>**`), keeping `**<pre>/<CLI blocks, file paths, the `ending_reason` enum, brand/tool tokens (KhelSutra/VLC/rally-annotator/MIT/GPU/LLM) and the VLC UI button labels** **\*\*literal**, and every honest caveat intact (`<title>/meta left English, consistent with A3). en authored ****byte-for-byte** (jsdom-extracted ? the A3 drift guard validates it); es/da/id en-seeded; ****kn/hi AI-drafted ? 3-lens adversarial verify (accuracy/fluency/integrity) ? revise** via a 10-agent workflow (16 kn + 10 hi findings folded), reusing the established kn/hi register; the shipped `requestAccess` translation preserved. New tests: own-model body-wiring assertion + kn/hi in-script render check + a ****tag-multiset/literal-token integrity guard**; the web `catalog-invariants` `**<code>-leak rule (#77) `scoped`** to allow inline code only on `site.ownModel.`. **\*\*Site 75 + web 215 green; web typecheck/coverage (93.4%, ? floor)/build green.** **\*\*Real-browser pass** (kn + hi render, `****` preserved, code blocks literal, **\*\*zero console errors**); screenshot `khelsutra-ownmodel-i18n-desktop.webm` + per-locale stills in `khelsutra-screenshot/`. kn/hi remain **\*\*native-review-pending** (the two human gates are unchanged). Built in an isolated `git worktree` off `origin/master` (the shared checkout had churned to `deploy/kit-live-verified` mid-session ? concurrency convention vindicated).

190

8. assistant (2026-06-22T01:56:07.306Z)

Both docs are thorough. Now let me read the LOCAL_APPLIANCE_ARCHITECTURE \$9 T-series, and cross-check the #77 native-review scope and watermark localization claims against git.

9. user (2026-06-22T01:56:08.251Z)

```
1      # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3      > **Status: `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
4      > **Owner: Avi . **Created: 2026-06-20 . **Last updated: 2026-06-21
5      > **Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
```

```

DONE)
5      > **Tracking anchors:** §9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this
repo; **consumes** the engine's decoupled `StorageBackend` + `python -m backend.worker` seams
and the FastAPI loop; **extends** the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha
screens. Engine-side changes here are **proposals to coordinate** with the active
decoupled/worker sessions ? not written unilaterally.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11     ## 0. TL;DR
12     **KhelSutra Studio** turns the engine from a CLI into a **single-box, UI-powered rally
indexer + reel generator**: the khelsutra `web/` React SPA, served behind an **edge auth gate**,
drives the **existing** engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on
**one machine**, with `StorageBackend=local` for the MVP. It is the **single-box deployment
profile of the just-shipped decoupled architecture** ? storage=local + compute=local is the
degenerate collapse of the same storage/worker split.
13
14     The **single most important caveat:** the engine has **zero app-layer auth and CORS
`***` `[verified backend/main.py:42]`, so safety is structural **only** because the engine stays
bound to `127.0.0.1:8000` `[verified backend/main.py:638]`, a firewall blocks inbound `8000`,
and a reverse proxy/tunnel is the sole authenticated ingress ? **the gate, not the app, is the
tenancy boundary.**
15
16     Two honest hardware truths: (1) the current droplet is **CPU-only** ? real detection
there is **Gemini (cloud)** or **stub (CPU mock)**; **real local-GPU detection needs a GPU
host** (native WASB, in-flight). (2) The owner priority is to **stand up a GPU host ASAP** ?
provider-agnostic (DO GPU droplet / GCE GPU VM / Cloud Run+GPU) ? but that runs **in parallel**
with the UI MVP, which ships on Gemini today.
17
18     ## 1. Problem & goal
19     The decoupled engine just shipped a `StorageBackend` ABC + a `python -m backend.worker
{infer|train}` dispatcher `[verified backend/storage/, backend/worker/__main__.py]`, and the
FastAPI loop already covers `process ? jobs ? query ? compile ? highlights` `[verified
backend/main.py]`. What's missing is a **console** that lets a non-CLI operator run the whole
loop on a single box, **honestly, behind a gate**.
20
21     **Headline CUJ (owner's words):** a **hosted website** where I pick an input ? **a
Google Drive file, a bucket file (`s3://`/`gs://`), or a local upload** ? it lands in the
**bucket (DO Spaces / GCS)** or locally ? I **generate highlights** on a **GPU-enabled host**
running the real e2e ? I **pick rallies, watch, and download** ? **completely from the UI.**
22
23     **Outcome ("good")**: an operator signs in, picks a source, watches it index, curates
rallies, downloads a reel ? on the **live CPU droplet today** (Gemini/stub), and on a **GPU host
later** (native WASB) with **no UI fork**.
24
25     **Read to get current:** `PER_RALLY_SELECTION.md` (§3/§4/§10 contract anchors), engine
`HOSTING_PLAN.md` (gating), `DECOUPLED_COMPUTE/HANDOFF.md` (what's built),
`VIDEO_LOCALITY_MODEL.md` (privacy/retention).
26
27     ## 2. Decisions locked
28     | # | Decision | Source / date | Implication |
29     |---|---|---|---|
30     | D1 | Reverse-proxied **single origin**; engine stays `127.0.0.1:8000` | `[verified
main.py:638]`; 2026-06-20 | CORS `` is moot from the browser ? the proxy is the boundary |
31     | D2 | **Edge auth**, zero engine auth code | `HOSTING_PLAN.md`; 2026-06-20 | Caddy
basicauth OR cloudflared + Cloudflare Access |
32     | D3 | MVP drives the **in-process FastAPI worker** loop, `storage=local` | `[verified
job_worker.py:37, local.py:33]`; 2026-06-20 | bucket worker = reserved scale path, out of MVP |
33     | D4 | Embed **RallyPicker** from `PER_RALLY_SELECTION.md` **verbatim** | PR [#11];
2026-06-20 | the console is a superset; do not redesign the picker |

```



```

34      | D5 | Console lives in the khelsutra **web/** React scaffold | `web/README.md`;
2026-06-20 | Cloudflare-Pages-style root, not engine static |
35      | D6 | Compute placement is a **setting**, rendered from `GET /api/capabilities` |
`[design]`; 2026-06-20 | UI offers ONLY what the box can honestly do |
36      | D7 | **Ingest is a source menu** (local upload · bucket URI · Drive) normalizing to
one video object | `[verified backend/storage/ref.py parse_uri]`; 2026-06-20 | Drive is a
**stub** today ? scoped honestly ($4.2) |
37      | D8 | GPU host is **provider-agnostic** (DO droplet / GCE VM / Cloud Run+GPU); storage
stays the bucket | Owner 2026-06-20 | compute and storage are separate planes; no provider lock-
in |
38      | D9 | **Observability is first-class** ? structured logs on every new path + a DoD item
| Owner 2026-06-20 | see $5; verified during CUJ replay |
39      | D10 | Delivery = a **self-hosted webserver + UI the user spins up** (NOT a desktop
app); the user points at **local storage + GPU/CPU OR cloud locations** and the UI drives the
tool | Owner 2026-06-20 | **supersedes `DESKTOP_APP_PLAN.md`**; one artifact (this appliance)
runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings |
40
41      ## 3. Foundation ? what the engine gives us today `[verified this session unless
noted]`
42      - **FastAPI loop:** `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ?
`POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}`
(FileResponse) `[main.py]`. **Chunked upload exists** ? `POST /api/upload/init` · `PUT
/api/upload/{id}/part/{i}` · `POST /api/upload/{id}/complete` · `DELETE /api/upload/{id}`
`[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` **never calls them** (it only
hits `/api/process` + `/api/compile` `[verified app.js:88,373]`). So browser upload is **client-
wiring only**, no engine change.
43      - **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)`
`[verified job_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the
clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
**separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
to serialize) ? it is **not** already 1.
44      - **Storage:** `storage.backend` default = `"local"` `[verified models.py:298]`;
`LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on
`src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-
local**. URI scheme `parse_uri` `[verified ref.py]`: bare `/local://` ? local; `s3://`
(AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **follow-up stub
(emulated only)**. Credentials never in config ? boto3 default chain / `~/aws` `[verified
s3.py]`.
45      - **Compute seam:** `indexing.detector_impl` ? `{auto=WSL, native=Linux-native,
`stub`=CPU mock}; default "auto" `[verified models.py:129]`, resolved in
`trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB is in-flight on branch
`feat/native-wasb-runner`:**`native_wasb_runner.py` **exists** `[verified]`;
**fusion_hybrid._build_native_runner` returns `NativeWasbRunner`** `[verified
fusion_hybrid.py:86]` while **`trajectory_hybrid._build_native_runner` still raises** `[verified
:57-63]`. Acceptance gate = **owner byte-parity** (WSL CSV == native CSV) on a real GPU box ?
not exercisable on the droplet or in the offline suite. Treat native as **not production-
verified**.
46      - **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ?
pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified
worker/__main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
`--set indexing.skip_ai_handoff=true` + `storage.s3`.
47      - **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
/api/health` **now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}`
`[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.
48      - **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts
the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the
full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md,
DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
there.
49      - **Absent (net-new):** `GET /api/capabilities`, `GET /api/videos` (only single-row
`get_video(id)`), `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any
**Dockerfile**, any per-video **cost ledger** `[verified ? grep returns none]`.
50
51      ## 4. Design / architecture

```

```

52
53     ### 4.1 Topology & request path
54     ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy /
static host** serving the `web/` SPA bundle and proxying `/api/` ? engine **FastAPI
`127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a
GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
site stays a **separate** systemd unit on a **separate** subdomain.
55
56     ` ` `
57     PUBLIC INTERNET
58     ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
59     ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
basicauth
60     ?

```

10. user (2026-06-22T01:56:09.780Z)

```

152:## 9. Deliverables & progress tracker ? **source of truth**
159:| T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key (P1)
| ? | owner | ? | ? |
160:| T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ? |
161:| T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `/api/upload/`
? server path ? `/api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
162:| T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163:| T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
167:| T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) | T4
| no | ? | ? |
168:| T11 | Retention sweep for `output/+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-*` + regenerable frames/staged/proxies; **reels/DB/CSVs
durable**; `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
169:| T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs
(M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` $9) | T4 | ? | ? | engine #280 + SPA #56 |

```

11. assistant (2026-06-22T01:56:13.862Z)

Let me see the full T-series table and the section around it, plus verify the #77 native-review scope (which catalogs it touched) and the watermark-localization claims (LW1/LW2/LW3 from memory) against khelsutra and engine git.

12. user (2026-06-22T01:56:15.352Z)

```

152     ## 9. Deliverables & progress tracker ? **source of truth**
153     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR
ships its tests + meaningful logs ($5).
154
155     | ID | Deliverable | Depends on | Gated? | Status | PR |
156     |----|-----|-----|-----|-----|----|
157     | T0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | owner | ? |
(this PR) |
158     | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |
159     | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160     | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
161     | T4 | S1 ingest: **local upload wired ?

```

```

([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `/api/upload/`
? server path ? `/api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
162 | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163 | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164 | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165 | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166 | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167 | T10 | Observability pass: correlation id end-to-end + structured logs + metrics (S5) |
T4 | no | ? | ? |
168 | T11 | Retention sweep for `output/`+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-*` + regenerable frames/staged/proxies; **reels/DB/CSVs
durable** ; `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
169 | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs
(M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` §9) | T4 | ? | ? | engine #280 + SPA #56 |
170 | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner
byte-parity (P4) | branch | ? owner | ? | ? |
171 | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
172
173 **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does **not** block on it (ships on Gemini).
174
175 **? Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).**
The alpha ships the **split** topology (SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel + CF Access over both ? the engine `HOSTING_PLAN.md` §2), **not** the single-
origin Caddy shell of **T2** (kept as the documented alternative ? §0/D1). The **deploy kit +
runbook** ([`deploy/DEPLOY_ALPHA.md`](../deploy/DEPLOY_ALPHA.md) +
`deploy/cloudflared.config.example.yml`) is shipped (? templates+runbook); the actual stand-up
(tunnel + Pages + CF Access + key rotation) is the **owner** step. The **T3**
raw-`:8000`-refused posture test is documented as the deploy kit's §6 verification. Enabled by:
engine exposure-safety boot posture (engine #294) + SPA `VITE_API_BASE` ([#64]).
176
177 ## 10. Open questions ? owner / external
178 **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:**
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.
179 **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming
endpoint; per-user scoping timing; containerization vs runbook.
180
181 ## 11. Definition of done
182 - [ ] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via
the ingest menu.
183 - [ ] Raw `:8000`/`:900x` unreachable from outside (**tested**, not assumed).
184 - [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped
honestly (not faked).
185 - [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
186 - [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via
`/api/videos` + `/api/reels`.

```

```

187 - [ ] **Observability:** correlation id threaded end-to-end; structured logs at every
new path; loud failures; logs inspected during CUJ replay.
188 - [ ] Retention sweep in place (**tool shipped** [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-schedule**
[engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301));
`GEMINI_API_KEY` rotated; `allowed_video_root` set.
189 - [ ] CI green (L2 contract + L4 regression); coverage ? floor; $9 tracker all
?/?-resolved.
190
191 ## 12. References
192 **Engine code `[verified this session]`:**
`backend/main.py:12-17,42,52,307-328,331-340,342-396,638`;
`backend/api/uploads.py:64,89,123,147`; `backend/api/job_worker.py:37`;
`backend/api/static/app.js:88,344,373`;
`backend/storage/{base.py,local.py:33,ref.py,___init___.py:43}`;
`backend/config/models.py:129,183,298`;
`backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86,
gemini.py:470-485, motion.py:191-198}`; `backend/pipeline/detectors/{native_wasb_runner.py,
stub_runner.py}`; `backend/worker/___main___.py:88,98,149`; `config.json` (`default_segmenter:
gemini`). **Engine docs:** `HOSTING_PLAN.md:42,102-108`; `VIDEO_LOCALITY_MODEL.md`;
`DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`;
`DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`;
`PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`,
`PORTABILITY.md`, `web/README.md`.
193
194 ---
195 ### Changelog
196 - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs
(FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin
spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor
re-verified against source before writing (catching the native-WASB partial-wiring +
`ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local),
multi-provider GPU, and first-class Observability directives.

```

13. user (2026-06-22T01:56:17.264Z)

```

=== #77 files ===
commit 9f1ec9c1d9ebeebe173fa5d727ec76d5555409a7
Author: Avi <avi.dullu@gmail.com>
Date: Mon Jun 22 00:46:09 2026 +0530

```

fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)

Adversarial translation audit of the AI-drafted Kannada/Hindi SPA console strings (B5/B6) surfaced real quality bugs in the non-site namespaces (app/bulk/grid/footer/result/ingest/errors). en (/locales/en) is the source of truth; the site.* marketing namespace (A3) is untouched. Fixes:

Correctness

- Brand stays Latin "KhelSutra" ? drop the Devanagari transliteration "???????" (hi: ingest.intro/reassure/fileHint) and the Kannada "?????????" (kn: same keys). A product name is never transliterated.
- footer.floorSomeRest (kn+hi): drop the leaked internal `min\_shot\_count` code token (kn even wrapped it in <code>) and the word "engine" ? kn "(?? ???? ???? ??????)", hi "(? ???? ???? ??)?", matching en's plain "(they're too short)".

Jargon leak ? en deliberately avoids "engine"/"server"/"bucket"; align kn/hi

- footer.floorOne, footer.floorAll: drop "engine"; rephrase passively ("it would be skipped" / "they'd all be skipped"), matching en.
- errors.networkError: "engine" ? the brand ("Can't reach KhelSutra?").
- ingest.intro, ingest.reassure: drop the bucket/engine/server clauses that were absent from en; match en's plainer meaning.
- ingest.hint, errors.invalidUri: en says "the machine running KhelSutra" (not "engine") ? align kn/hi; ingest.hint also "bucket link" ? "cloud-

```
storage link". (ingest.uriAriaLabel keeps "engine host"/"bucket URL"
because en itself uses them there.)
=== #86 own-model ===
commit 17242a9937bd3b3083a6ef2461c311c5a44b3092
Author: Avi <avi.dullu@gmail.com>
Date: Mon Jun 22 07:21:22 2026 +0530
```

feat(site): localize own-model body content to kn/hi (#86)

* feat(site): localize own-model body content to kn/hi

The developer/power-user "own your model" page (site/public/own-model.html) was English-only on kn/hi ? only its switcher + Request-access link were localized. Wire its full body (hero, annotate/train/setup steps, moat grid, honest-framing fences, "what this is not", CTA, footer) through the i18n shim: 52 new site.ownModel.* keys via data-i18n / data-i18n-html (preserving inline /<i>/<a>/<code>/). CLI/<pre> code blocks, file paths, the ending_reason enum, brand/tool tokens (KhelSutra/VLC/rally-annotator/MIT/GPU/LLM) and the VLC UI button labels stay literal; the honest caveats are kept intact.

- en authored byte-for-byte (jsdom-extracted), validated by the A3 drift guard
- es/da/id en-seeded; kn/hi AI-drafted -> 3-lens adversarial verify (accuracy/fluency/integrity) -> revise (16 kn + 10 hi findings folded);

14. assistant (2026-06-22T01:56:22.700Z)

Now I understand the #77 scope precisely: it covered the **kn/hi SPA console** (non-site namespaces: app/bulk/grid/footer/result/ingest/errors), explicitly NOT touching `site.\*` (A3 marketing) and NOT es/da/id. Let me verify the watermark-localization claims (memory says LW1 engine #304 / LW2 khelsutra #73 / LW3 engine #306) and confirm what remains for native review. Let me check the engine origin/master and khelsutra #73.

15. user (2026-06-22T01:56:24.784Z)

```
=== #73 LW2 ===
commit c1848df27fd87e44bf7043f0ddd0b703186121d3
Author: Avi <avi.dullu@gmail.com>
Date: Sun Jun 21 23:55:33 2026 +0530
```

feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)

The SPA now records the language it was used in on the two engine calls that need it:

- compileReel + processVideo (api/client.ts) take an optional `locale`, included in the body and omitted when absent (JSON.stringify drops undefined ? full back-compat).
- IngestPanel passes i18n.resolvedLanguage on both ingest paths (URI + file upload),

threaded

through useIngestJob.submit(uri, locale) ? processVideo.

- App.onBuild passes i18n.resolvedLanguage to compileReel ? the engine localizes the reel's branding watermark (engine LW1) to the user's language.

Tests: api/locale.test.ts (locale included when present, omitted when absent, on both calls)

+

a build-error.test App-integration check (the active locale reaches /api/compile). 177 web tests

green; tsc + build clean. Inert/back-compat until the engine LW1 (watermark) + LW3 (analytics) land.

Co-authored-by: Claude Opus 4.8 <noreply@anthropic.com>

```
web/src/App.tsx | 5 ++--
web/src/api/client.ts | 11 +++++--
web/src/api/locale.test.ts | 45 ++++++
web/src/components/IngestPanel.tsx | 8 +++--
web/src/components/build-error.test.tsx | 20 ++++++

```

16. user (2026-06-22T01:56:26.497Z)

```
=== engine log (watermark/locale) ===
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
2afbadf fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project
(#275)
=== full recent ===
a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers
(#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-
OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export)
(default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place)
(default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the
bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-
OFF) (#302)
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities,
/api/videos, /api/reels (#300)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state
(default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI
lanes (#297)
3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard
(default-OFF) (#296)
fc78fce feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config
(#295)
0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed
engine (default-OFF) (#294)
```

17. assistant (2026-06-22T01:56:31.766Z)

Confirmed. The watermark localization chain is fully shipped on engine origin/master: LW1 (#304, watermark localized + OFL Noto fonts), LW3 analytics (#306, v7 DB locale per job), and khelsutra LW2 (#73, SPA sends locale). So the watermark strings ARE now an AI-drafted localized surface that needs native review.

Let me verify exactly which catalogs exist (es/da/id + watermark strings location) and confirm the native-review scope precisely ? what's been reviewed vs not. Let me check the watermark string source in the engine.

18. user (2026-06-22T01:56:33.902Z)

```
=== locales dirs ===
da
en
es
hi
id
kn
=== es/da/id catalog status (are they translated or en-seeded?) ===
17242a9 feat(site): localize own-model body content to kn/hi (#86)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero
(#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
```

19. user (2026-06-22T01:56:36.730Z)

=== watermark localized strings in engine origin/master ===

--- where are the localized watermark strings? ---

tree origin/master:backend/pipeline

audio/

detectors/

personalization/

segmenters/

stitcher/

validators/

origin/master:backend/pipeline/stitcher/watermark_i18n.py:3:The SPA passes the UI locale to ``POST /api/compile``; this maps it to a localized branding string

origin/master:backend/pipeline/stitcher/watermark_i18n.py:4:and a script-appropriate bundled font, so a Kannada/Hindi/? user's reel carries the watermark in

origin/master:backend/pipeline/stitcher/watermark_i18n.py:5:their language. The brand token ("KhelSutra.guru") stays Latin; only the surrounding phrase localizes.

origin/master:backend/pipeline/stitcher/watermark_i18n.py:8: - en/es/da/id (Latin) ? the configured default (bundled RobotoSlab, Apache-2.0)

origin/master:backend/pipeline/stitcher/watermark_i18n.py:9: - kn ? Noto Sans Kannada · hi ? Noto Sans Devanagari (OFL-1.1, bundled ? the D7 OFL-for-media

origin/master:backend/pipeline/stitcher/watermark_i18n.py:10: carve-out;

FRIENDLY_AND_MULTILINGUAL \$4 anticipated this).

origin/master:backend/pipeline/stitcher/watermark_i18n.py:12:Honest fallbacks: an

unknown/missing locale (or a locale with no entry, incl. ``en``) returns the

origin/master:backend/pipeline/stitcher/watermark_i18n.py:13:watermark UNCHANGED, so

existing/default behavior is preserved. If a script font file is somehow

origin/master:backend/pipeline/stitcher/watermark_i18n.py:15:and the compiler already retries

the encode without the watermark if drawtext can't render, so a

origin/master:backend/pipeline/stitcher/watermark_i18n.py:16:font gap can never nuke the reel.

Kept config-package-agnostic (operates on a dict).

20. assistant (2026-06-22T01:56:40.394Z)

The watermark localized strings live in `backend/pipeline/stitcher/watermark\_i18n.py` on engine origin/master. Let me see the actual localized phrases to confirm which languages have AI-drafted text needing native review.

21. user (2026-06-22T01:56:41.712Z)

```
21:# This module is backend/pipeline/stitcher/ ? parents[2] == backend/ (matches compiler.py).
```

```
22:_FONTS_DIR = Path(__file__).resolve().parents[2] / "assets" / "fonts"
```

```
26:WATERMARK_TEXT: Dict[str, str] = {
```

```
27:     "es": "Generado con KhelSutra.guru",
```

```
28:     "da": "Lavet med KhelSutra.guru",
```

```
29:     "id": "Dibuat dengan KhelSutra.guru",
```

```
30:     "kn": "KhelSutra.guru ??? ????",
```

```
31:     "hi": "KhelSutra.guru ?? ????",
```

```
36:FONT_FILE_BY_LOCALE: Dict[str, str] = {
```

```
37:     "kn": "NotoSansKannada-Variable.ttf",
```

```
38:     "hi": "NotoSansDevanagari-Variable.ttf",
```

```
54:     base = _base_locale(locale)
```

```
55:     if base not in WATERMARK_TEXT:
```

```
59:         wm: Dict[str, Any] = watermark.model_dump()
```

```
61:         wm = dict(watermark)
```

```
65:         wm["text"] = WATERMARK_TEXT[base]
```

```
66:         font_file = FONT_FILE_BY_LOCALE.get(base)
```

```
68:         font_path = _FONTS_DIR / font_file
```

```
70:         wm["font_path"] = str(font_path)
```

22. assistant (2026-06-22T01:56:46.475Z)

Now I have a complete, ground-truthed picture. The watermark localized phrases (es/da/id/kn/hi)

are all AI-drafted and need native review ? they are a separate translated surface beyond the SPA console (#77) and the site marketing (A3).

Let me do a final cross-check on the §8 owner-gated Phase B/C status and confirm es/da/id catalogs are AI-drafted (not en-seeded), then I have everything.

23. user (2026-06-22T01:56:48.598Z)

=== es app.title (translated or en?) ===

```
{
  "app": {
    "badge": "elige tus jugadas · alfa",
    "title": "Elige las jugadas para tu reel",
    "intro": "Filtra por <b>duración de la jugada</b> ? p. ej. <i>?las 15 más largas?</i> ? y
luego ajusta tocando las tarjetas. Todo empieza <b>seleccionado</b>; solo quita las que no
quieras."
  },
  "lang": {
    "label": "Idioma",
    "en": "English",
    "kn": "?????",

```

=== da ===

```
{
  "app": {
    "badge": "vælg dine dueller · alfa",
    "title": "Vælg duellerne til din reel",
    "intro": "Filtrér efter <b>duellens længde</b> ? f.eks. <i>?de 15 længste?</i> ? og

```

=== id ===

```
{
  "app": {
    "badge": "pilih reli kamu · alfa",
    "title": "Pilih reli untuk reel kamu",
    "intro": "Filter berdasarkan <b>durasi reli</b> ? mis. <i>?15 terpanjang?</i> ? lalu sesuaik

```

24. assistant (2026-06-22T01:56:59.951Z)

Confirmed: es/da/id catalogs are fully AI-translated (not en-seeded) ? so they need native-speaker review for the SPA. I now have a complete, git-grounded picture. Let me compile the structured output.

Summary of ground-truth findings:

- **§7 FRIENDLY tracker: all A/B rows ? except B7** (engine backend l10n, owner-gated ?). A2 shipped (#82), contradicting two stale "A2 deferred out of v1" mentions in the doc's §7 milestone prose.
- **SPA = 6 languages** (en/kn/hi via FRIENDLY B1-B6; es/da/id via MULTILINGUAL Phase A #67). Site = en/kn/hi.
- **Watermark localization fully shipped** (engine #304/#306 + khelsutra #73) ? the es/da/id/kn/hi watermark phrases in `watermark\_i18n.py` are AI-drafted, native-review-pending.
- **Native review done so far: only the kn/hi SPA console (#77)**. NOT reviewed: es/da/id SPA catalogs (full), kn/hi + es/da/id site marketing (A3/#74, #86), the watermark phrases.
- **Two human DoD gates open** (FRIENDLY §9): native-speaker review of remaining AI-drafted catalogs + owner font-glyph eyeball.
- **MULTILINGUAL Phase B (Thai)/C (CJK) owner-gated** ? §8 Q1-Q6 unanswered (E0 approval gate not cleared for B/C).
- **T-series**: T2 deploy effectively done (split topology, per context); genuinely-open non-gated UX-adjacent items are T5 (progress polling/restart-failure UX) and T6 (embed RallyPicker) ? but those are appliance/SPA-functionality, lower priority than the human launch gates. T13/T14 owner-gated.

25. user (2026-06-22T01:57:43.200Z)

Output does not match required schema: root: must have required property 'next_items'

26. assistant (2026-06-22T01:57:46.529Z)

The `next\_items` was sent as a string instead of being part of the structured object. Let me resubmit with the correct shape.

27. user (2026-06-22T01:58:14.083Z)

Structured output provided successfully

28. assistant (2026-06-22T01:58:18.345Z)

Done. Structured output submitted with git-grounded status and prioritized next items for the khelsutra UX/localization/runnable-tool track.